Naive Bayes.

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Training data:
CSK

toss	Venue	Outlook	Result.
won	Mumbai	Overcast	won.
lost	~~Mumbai~~ Chennai	Sunny	~~lost~~ won
won	Kolkata	Sunny	won.
won	Chennai	Sunny	won.
lost	Mumbai	Sunny	lost
won	Chennai	Overcast	lost.
won	Kolkata	Overcast	lost
won	mumbai	Sunny	won.

Test: {lost, Mumbai, Sunny}. (Here we can use $\cap$, comma, & anyone.)

$P(W \mid \text{Lost} \cap \text{Mumbai} \cap \text{Sunny})$

$$= \frac{P(\text{Lost} \cap \text{Mumbai} \cap \text{Sunny} \mid W) \cdot P(W)}{P(\text{Lost, Mumbai, Sunny})}$$

[assume = 0.56]

$P(L \mid \text{Lost, Mumbai, Sunny})$

$$= \frac{P(\text{Lost, Mumbai, Sunny} \mid L) \cdot P(L)}{P(\text{Lost, Mumbai, Sunny})}$$

[assume = 0.423]

Which is higher that will happen. Here then win.

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Here; Denominator : same [P(lost, Mumbai, Sunny)], so; ignore it coz it won't help you for answer.

$$P(W \mid \text{Lost, Mumbai, Sunny}) = \underbrace{P(\text{lost, mumbai, Sunny} \mid W)}_{\text{II}} \cdot \underset{5/8}{P(W)}$$

$$= P(\text{lost} \mid W)\, P(\text{Mumbai} \mid W) \cdot P(\text{Sunny} \mid W) \cdot \frac{5}{8}$$

$$= \frac{1}{5} \cdot \frac{2}{5} \cdot \frac{4}{5} \cdot \frac{5}{8}$$

$$= 0.3 \quad \text{(assume not calc)}.$$

Do same with

$$P(L \mid \text{Lost, Mumbai, Sunny}) \approx 0.2$$

Here. $0.3 > 0.3$

Hence, win

Naive Bayes Mathematical Intuition:

~~Chain Rule~~

$$\{x_1 \cap x_2 \cap x_3 \cdots \cap x_n\}.$$

$$x \rightarrow \{x_1, x_2, x_3, \ldots, x_n\}.$$

$$c = \{c_1, c_2, c_3, \ldots, c_k\}.$$

Conditions, c_1 = win, c_2 = loss.

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Now, chain rule for conditional Probability.

$$P(C_k \mid x) = \frac{P(x \mid C_k) \cdot P(C_k)}{P(x)}$$

remain common so,

$$P(C_k \mid x) = P(x \mid C_k) \cdot P(C_k)$$

$$= \frac{P(x \cap C_k)}{P(C_k)} \cdot P(C_k)$$

$$= P(x \cap C_k)$$ replacing $\cap$ by comma ;

$$= P(x, C_k)$$

$$= P(\boxed{x_1}, \boxed{x_2, x_3, \ldots x_n, C_k})$$

~~Let A ∩ B~~

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$= P(x_1, x_2, x_3, x_4, \ldots x_n, C_k)$$

$$= \underbrace{P(x_1 \mid x_2, x_3, \ldots, x_n, C_k)} \cdot P(x_2, x_3 \ldots x_n, C_k)$$

or, $P(C_k \mid x) = a \cdot P(\boxed{x_2}, \boxed{x_3, \ldots x_n, C_k})$

$$= a \cdot \underbrace{P(x_2 \mid x_3, x_4, \ldots x_n, C_k)} \cdot P(x_3, x_4, \ldots x_n, C_k)$$

$$= ab\, P(x_3, x_4, \ldots, x_n, C_k)$$

$$= \cdot \cancel{P(x_n \mid C_k) \cdot P(C_k)}$$

$$= ab\, P(x_n \mid C_k) \cdot P(C_k)$$

$$\therefore P(C_k \mid x) = P(x_1 \mid x_2, x_3 \ldots, x_n, C_k) \cdot P(x_2 \mid x_3, \ldots x_n, C_k) \ldots P(x_{n-1} \mid x_n, C_k) \cdot P(x_n \mid C_k) \cdot P(C_k)$$